Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int n;
    int count = 0;
    int c = 0;

    scanf("%d", &n);

    // Initial statement: c = 0
    count++;

    // for(int i = n/2; i < n; i++)
    for (int i = n / 2; count++, i < n; i++) {
        // for(int j = 1; j < n; j = 2 * j)
        for (int j = 1; count++, j < n; j = 2 * j) {
            // for(int k = 1; k < n; k = k * 2)
            for (int k = 1; count++, k < n; k = k * 2) {
                c++;
                count++;
            }
        }
    }

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.